Complex centered moments below two are not strongly Leibniz

Candidate counterexample packet for arXiv:1409.1207

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Outcome. The literal complex-valued question has a negative answer. For every $1 \leq p < 2$, the centered L^p seminorm fails the strong inverse inequality already on the uniform three-point probability space. A rational counterexample at $p = 1$ is

$$f = \left(1, \frac{24 - 7i}{25}, \frac{26 + 7i}{25}\right).$$

1. The question and the exact scope

For a probability space and $1 \leq p < \infty$, the source sets

$$\sigma_p(f) = \|f - \mathbb{E}f\|_p.$$

A seminorm is strongly Leibniz only if every invertible f satisfies

$$\sigma_p(f^{-1}) \leq \|f^{-1}\|_\infty^2 \sigma_p(f). \tag{1}$$

The source defines f to be complex-valued, proves special real cases, and leaves open whether all centered moments on general probability spaces are strongly Leibniz.

standard deviation. After that we shall present some scattered Leibniz-type result for different moments on different (discrete, general) probability spaces. We leave open the question whether all centered moments in general probability spaces define a strongly Leibniz seminorm. Lastly, in Section 3, we shall answer affirmatively Rieffel's question on the standard deviation in non-commutative probability spaces.

Figure 1: The broad strong-Leibniz question, source PDF page 2.

Theorem 1 (Three points suffice). *Fix $1 \leq p < 2$. On $\mathbb{C}^3$ with the uniform probability measure there is an invertible f for which*

$$\sigma_p(f^{-1}) > \|f^{-1}\|_\infty^2 \sigma_p(f).$$

Thus σ_p is not strongly Leibniz on complex-valued probability spaces for any $1 \leq p < 2$.

Proof intuition. Put three complex numbers around the point 1 so that two displacements cancel exactly. Inversion bends this almost-linear configuration. The two large centered inverse displacements lose only order k^{p-2} in pth power, while inversion creates a third displacement whose contribution is order one after the common scaling is removed. Below $p = 2$ the loss vanishes and that extra contribution forces the inverse seminorm upward.

2. An exact one-parameter family

For $k > 0$, set

$$z_k = \frac{k^2 - 1 - 2ki}{k^2 + 1}, \qquad f_k = (1, z_k, 2 - z_k).$$

We have $|z_k| = 1$ and

$$|2 - z_k|^2 = \frac{k^2 + 9}{k^2 + 1} > 1.$$

Consequently f_k is invertible and

$$\|f_k^{-1}\|_\infty = 1. \tag{2}$$

Also $\mathbb{E}f_k = 1$ and

$$|z_k - 1| = |1 - z_k| = \frac{2}{\sqrt{k^2+1}},$$

so

$$\sigma_p(f_k)^p = \frac{2^{p+1}}{3(k^2+1)^{p/2}}. \tag{3}$$

Write $a_k = f_k^{-1}$ and $D_k = (k^2+1)(k^2+9)$. Directly,

$$a_k = \left(1, \frac{k^2-1+2ki}{k^2+1}, \frac{k^2+3-2ki}{k^2+9}\right).$$

Subtracting its coordinate mean and simplifying gives the three moduli

$$\frac{8}{3\sqrt{D_k}}, \qquad \frac{2\sqrt{9k^2+25}}{3\sqrt{D_k}}, \qquad \frac{2\sqrt{9k^2+1}}{3\sqrt{D_k}}. \tag{4}$$

Equations (3)–(4) yield the exact ratio

$$\left(\frac{\sigma_p(f_k^{-1})}{\sigma_p(f_k)}\right)^p = \frac{4^p + (9k^2+25)^{p/2} + (9k^2+1)^{p/2}}{2\,3^p(k^2+9)^{p/2}}. \tag{5}$$

3. Why every exponent below two fails

Let $q = p/2 < 1$. The numerator minus the denominator in (5) is

$$F_p(k) = 4^p + (9k^2+25)^q + (9k^2+1)^q - 2(9k^2+81)^q. \tag{6}$$

For either $c = 25$ or $c = 1$, the mean value theorem gives

$$(9k^2+c)^q - (9k^2+81)^q = O(k^{2q-2}) = O(k^{p-2}).$$

Since $p < 2$, both differences tend to zero. Therefore

$$\lim_{k\to\infty} F_p(k) = 4^p > 0.$$

For every fixed $p < 2$, all sufficiently large k make the ratio in (5) strictly larger than one. Together with (2), this violates (1) and proves Theorem 1.

4. A rational counterexample at $p = 1$

Take $k = 7$. Then

$$f = \left(1, \frac{24-7i}{25}, \frac{26+7i}{25}\right), \qquad f^{-1} = \left(1, \frac{24+7i}{25}, \frac{26-7i}{29}\right),$$

and $\|f^{-1}\|_\infty = 1$. Since $\mathbb{E}f = 1$,

$$\sigma_1(f) = \frac{2\sqrt{2}}{15}. \tag{7}$$

The mean of f^{-1} is $(2071+28i)/2175$. The moduli of its three centered coordinates are

$$\frac{4\sqrt{29}}{435}, \qquad \frac{\sqrt{13514}}{435}, \qquad \frac{\sqrt{12818}}{435}.$$

Hence

$$\sigma_1(f^{-1}) = \frac{4\sqrt{29} + \sqrt{12818} + \sqrt{13514}}{1305}. \tag{8}$$

The strict comparison needs no decimal approximation:

$$4\sqrt{29} + \sqrt{12818} + \sqrt{13514} > 4\cdot 5 + 113 + 116 = 249,$$

whereas

$$174\sqrt{2} < 174\frac{71}{50} = \frac{6177}{25} < 249.$$

Thus the numerator in (8) exceeds $174\sqrt{2}$, which is exactly 1305 times (7). This proves $\sigma_1(f^{-1}) > \sigma_1(f)$.

5. Boundary and literature status

The counterexample is complex-valued. It does not contradict the source's real results on at most four atoms. The later paper arXiv:1601.00440 proves the ordinary Leibniz product inequality for real centered L^p moments, but explicitly leaves their strong inverse property open. The present argument does not decide that real conjecture, nor complex exponents $p > 2$. At $p = 2$, standard deviation is known to be strongly Leibniz, and (5) is strictly below one, consistently with that theorem.

Upgrade and novelty audit. Eight focused stages were completed: exact-scope and later-literature audits; finite inverse search; complex majorization obstruction; optimized witness; rationalization; exact family calculation; and the all-$p < 2$ asymptotic upgrade. Bounded primary-source searches through 17 August 2026 found no paper recording this complex three-point obstruction. Priority nevertheless requires broader expert review.

References

[1] A. Besenyei and Z. Léka, *Leibniz seminorms in probability spaces*, J. Math. Anal. Appl. 429 (2015), 1178–1189; arXiv:1409.1207.

[2] Z. Léka, *Symmetric seminorms and the Leibniz property*, J. Math. Anal. Appl. 451 (2017), 682–704; arXiv:1601.00440.